

Pankaj Maulekhi

AI/ML Engineer — Generative AI & Agentic Systems

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Professional Summary

AI/Data Science undergraduate at IIITDM Kurnool with a focus on generative modeling and agentic AI systems. Built and published PM-MiniFinLLM, a 110M-parameter finance LLM pretrained from scratch on 2.36B tokens with SFT and DPO alignment.. Core skills include LLM training and fine-tuning, and image generation models (diffusion, GANs). Experienced in MLOps and cloud deployment using AWS (SageMaker, ECR, EC2) , Docker, and CI/CD pipelines, with a focus on building production-grade generative AI systems.

Technical Skills

- **Languages:** Python, C++, C, SQL
- **Data Analysis and Visualization:** Pandas, NumPy, Matplotlib, Plotly, Seaborn, BeautifulSoup
- **Machine Learning:** Scikit-Learn, XGBoost, LightGBM, Feature Engineering, Hyperparameter Tuning
- **Deep Learning:** PyTorch, TensorFlow (Keras), Transformers , Computer Vision, NLP, Distributed Training (DDP)
- **Generative AI:** Diffusion Models, Flow Matching, GANs , Hugging Face, Fine-Tuning, Quantisation
- **Agentic AI and RAG:** LangChain, LangGraph, LangSmith, MCP, RAG, Vector Databases, Agentic Pipelines, Ragas
- **MLOps and Deployment:** MLflow, DVC, Optuna, FastAPI, Docker, GitHub Actions (CI/CD), AWS (S3, EC2, ECR, SageMaker, Bedrock)

Education

Period	Institution / Board	Degree / Course	Marks / Grade
2024 – Present	IIITDM Kurnool	B.Tech (AI and Data Science)	CGPA 8.1 (3 rd Sem)
2024	Jawahar Navodaya Vidyalaya Almora (CBSE)	Class XII	95.4%
2022	Jawahar Navodaya Vidyalaya Almora (CBSE)	Class X	96.4%

Achievements

- Achieved **98.38 percentile** in JEE Mains 2024, competing among over 1.4 million candidates nationwide.
- Solved **400+** problems on LeetCode, covering Data Structures, Algorithms, and Dynamic Programming.

Projects

- **PM-MiniFinLLM — Finance LLM from Scratch**  [GitHub](#)
Tech: PyTorch, HuggingFace Transformers, TRL, DPO, PyTorch DDP, FP16 Mixed Precision
 - Built a 110M-parameter GPT-2 style decoder-only transformer with RoPE positional embeddings, covering the full LLM pipeline from pre-training to alignment
 - Pre-trained on 2.36B tokens from SEC EDGAR filings using Distributed Data Parallel (DDP) training across 2× NVIDIA T4 GPUs with FP16 mixed precision
 - Fine-tuned on 28K financial QA samples via HuggingFace SFTTrainer and aligned using a custom DPO training loop; publicly registered on HuggingFace Hub
- **Neural Style Transfer System**  [GitHub](#)
Tech: TensorFlow, OpenCV, ONNX, MLflow, GitHub Actions, FastAPI, Docker, AWS (S3, EC2, ECR)
 - Built an AdaIN-based neural style transfer model from scratch on PASCAL VOC with custom loss functions, enabling real-time stylization with arbitrary reference images
 - Exported the trained model to ONNX for optimized inference and deployed a scalable REST API via FastAPI and Docker on AWS (S3, EC2, ECR)
 - Automated the full deployment pipeline with GitHub Actions CI/CD and MLflow experiment tracking for reproducible training runs
- **Self-RAG & Corrective RAG for the Indian Constitution & IPC**  [GitHub](#)
Tech: LangGraph, LangChain, DeepSeek R1 & V3, AWS Bedrock, ChromaDB, Sentence-Transformers, Ragas, Arize Phoenix
 - Designed a 13-node LangGraph state machine implementing Self-RAG and Corrective RAG (CRAG) for hallucination-resistant QA over the Constitution of India and the IPC, with DeepSeek R1 chain-of-thought query routing across retrieval, web search, and direct generation
 - Built parallel multi-query vector retrieval with fan-out/fan-in via LangGraph’s Send API and Map-Reduce relevance filtering, backed by ChromaDB and Sentence-Transformers embeddings, with automatic Tavily web search fallback
 - Implemented multi-stage self-correction (grounding checker + relevance judge) and evaluated via a custom Ragas pipeline, achieving 0.96 faithfulness, 0.95 context precision, and 0.90 context recall
- **Resume Tailor Agent**  [GitHub](#)
Tech: Python, LangGraph, LangChain, FastAPI, Model Context Protocol (MCP), MySQL
 - Developed a multi-agent LangGraph workflow that parses job descriptions and dynamically selects and rewrites the top 3 most relevant projects from a portfolio pool to maximize ATS alignment
 - Integrated a custom Model Context Protocol (MCP) server for automated LaTeX reading and PDF compilation, backed by a MySQL database for task tracking
 - Built a FastAPI backend with a Chrome Extension frontend, delivering an end-to-end automated document-optimization pipeline